

services for experimentation and low-volume workloads, self-hosted for high-volume production.

Model performance degradation:

Implement continuous monitoring with automated alerts on accuracy drops. Establish feedback loops capturing production errors for retraining. Schedule regular model refresh cycles—monthly or quarterly, depending on domain volatility. Maintain shadow deployments of candidate models to enable rapid switching when primary models degrade. Treat model operations as ongoing programmes, not one-time projects.

Organisational resistance: Design for human-in-the-loop workflows initially. Present AI outputs as recommendations requiring approval rather than automated actions. Provide transparency into model reasoning through attention visualisation or chain-of-thought outputs. Measure and communicate impact in business terms: time saved, revenue generated, errors prevented. Build trust through consistent performance and clear escalation paths when the system encounters edge cases.

Security and compliance concerns:

Engage security and compliance stakeholders from day one. Deploy in secure environments with appropriate network isolation, encryption at rest and in transit, and access controls. Implement comprehensive audit logging. For sensitive data, use on-premises or VPC deployments with no data egress. Document data handling procedures, model training provenance, and decision-making processes. Many organisations find that local SLM deployments actually simplify compliance compared to third-party API usage.

Best practices for SLM deployment

Organisations operating SLMs at scale converge on similar operational patterns. These practices separate experimental deployments from business-critical systems.

Resist the temptation to build general-purpose AI platforms. Begin with a single, well-defined use case with clear success metrics. Prove value. Then expand systematically to adjacent use cases, leveraging shared infrastructure and learned expertise. Organisations attempting to boil the ocean with universal AI initiatives consistently underdeliver compared to those pursuing focused, sequential deployments.

The performance ceiling is determined by training data quality, not model size. Allocate more resources to data curation, cleaning, and validation than to infrastructure or compute. Establish clear data quality standards. Review training examples manually. Remove ambiguous or contradictory data. This unglamorous work drives outsized performance improvements.

Modern quantization techniques enable 4-bit or 8-bit inference with minimal accuracy loss, reducing

memory requirements by 4-8x and increasing throughput proportionally. Every SLM deployment should evaluate quantized variants. The performance-efficiency tradeoff typically favors quantization except for the most demanding tasks. This single optimisation can eliminate 75% of infrastructure costs.

Automated evaluation against golden datasets should run on every code commit, model update, and production deployment. Maintain diverse test suites covering common cases, edge cases, adversarial inputs, and business-critical scenarios. Track metrics over time to detect gradual degradation. Production systems should continuously evaluate a sample of outputs against human judgment, feeding this data back into training pipelines.

Assume your current model will be replaced within 6-12 months. Abstract model interfaces to enable swapping base models without application

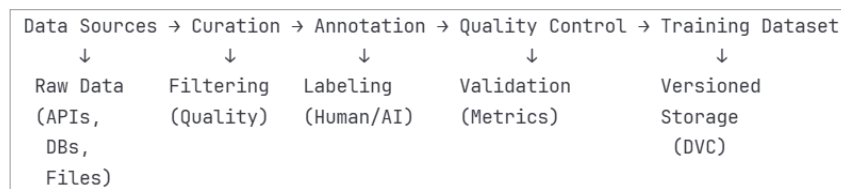


Figure 2: Data pipeline architecture

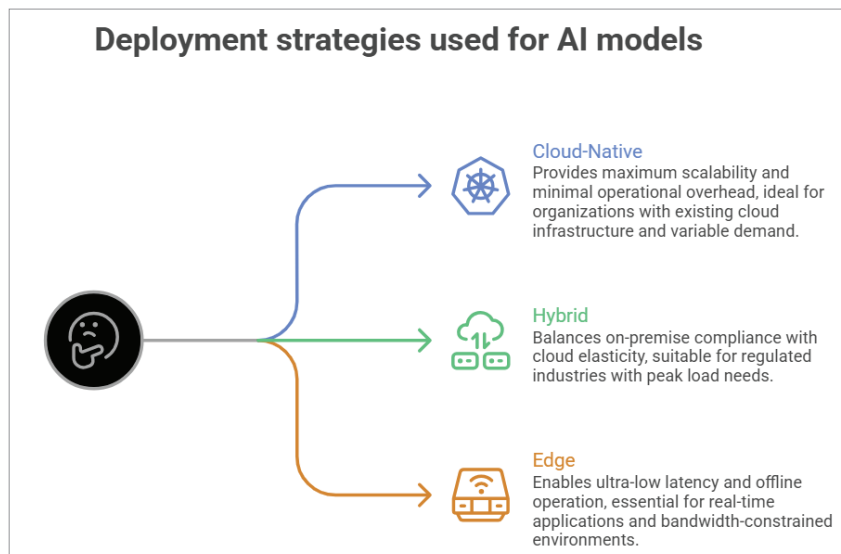


Figure 3: Integration patterns for enterprise environments